

FIELD VISIT REPORT

Tea Research Institute of Sri Lanka, Talawakele

Design of Automation for Hot and Ambient Air Mixing System & Supply System
for Efficient Tea Withering — ME325 Design Project

Date of Visit	10 June 2026
Location	Tea Research Institute (TRI), Talawakele
Project Team	WitherMinds — Dakshani, Shanuka, Savini
Supervisor	Dr. I. W. Kularathne
Course	ME325 — Mid-Term Design Project
Purpose of Visit	Field data collection from an operational tea withering trough to characterise existing air-mixing and airflow-distribution performance

1. Introduction

As part of the ME325 design project “Design of Automation for Hot and Ambient Air Mixing System & Supply System for Efficient Tea Withering,” the WitherMinds team requires field measurements from a working withering trough to validate the CFD model of the existing air-supply system and to inform the design of the proposed automated mixing and PLC-based control system.

On 10 June 2026, the team visited the Tea Research Institute (TRI) of Sri Lanka in Talawakele to record temperature, airflow, and pressure data from an operating withering trough under several hot-air / cold-air damper settings. This report documents the purpose, methodology, and data obtained during that visit, and how the results feed into the ongoing project.

2. Objectives of the Field Visit

- Record wet-bulb and dry-bulb temperatures at the top, middle, and bottom of the withering trough at multiple trial (damper) conditions.
- Measure air-flow readings across the cold-air and hot-air duct cross-sections to characterise mixing uniformity.
- Record the dry-bulb air temperature at increasing distances along the trough (9 ft, 26 ft, and 46 ft from the air-inlet pan) to capture the temperature gradient along the withering bed.
- Measure total and static air pressure at points along the trough to assess pressure drop along the airflow path.
- Use the above data set to validate the project's CFD model of the existing system and identify hotspots / uneven flow regions that the proposed automated mixing system must address.

3. Instrumentation & Measurement Methodology

Measurements were taken over five trials between approximately 6:00 PM and 10:30 PM, each trial corresponding to a different cold-air / hot-air damper opening combination (summarised in Section 4.1). For each trial, the following four measurement sets were recorded:

3.1 Trough Temperature (Wet Bulb / Dry Bulb)

A wet-bulb/dry-bulb hygrometer pair was used at three lateral positions (1, 2, 3) across the trough width, at three vertical levels — top, middle, and bottom of the leaf bed — giving nine temperature readings per bulb type per trial.

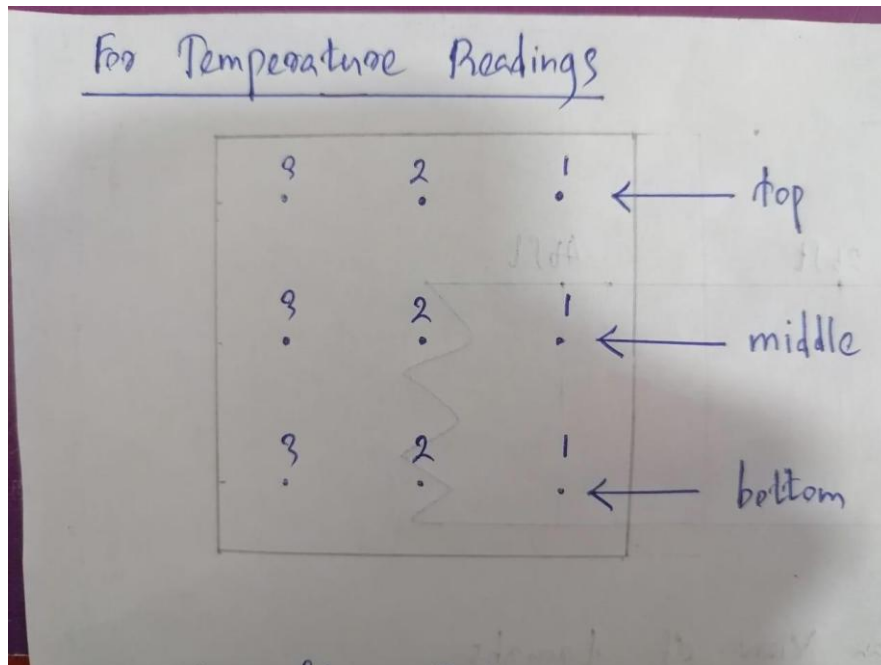


Figure 1: Layout of temperature measurement points (top, middle, bottom × positions 1–3) across the trough cross-section.

3.2 Air Flow Rate

Air-flow readings (FRM) were taken across a 3×3 grid on the cold-air damper face, and at four points across the hot-air duct, to evaluate the uniformity of air distribution entering the trough. The cold-damper face area was measured as 3.3 ft × 2.1 ft = 6.93 ft², used to convert local flow readings to volumetric flow (Flow × Area = Volumetric Air Flow).

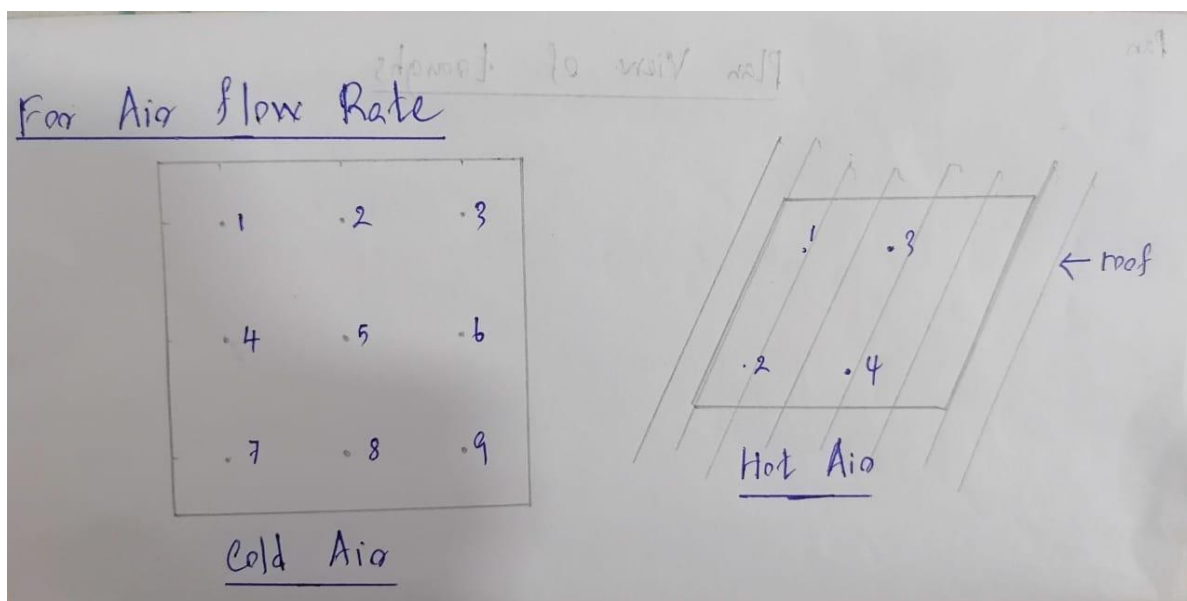


Figure 2: Grid of air-flow measurement points on the cold-air damper face (left) and hot-air duct (right).

3.3 Dry-Bulb Temperature Along the Trough

A dry-bulb thermometer was moved along the length of the trough and readings were logged at the air-inlet pan and at 9 ft, 26 ft, and 46 ft downstream, to capture how withering-air temperature decays with distance from the plenum.

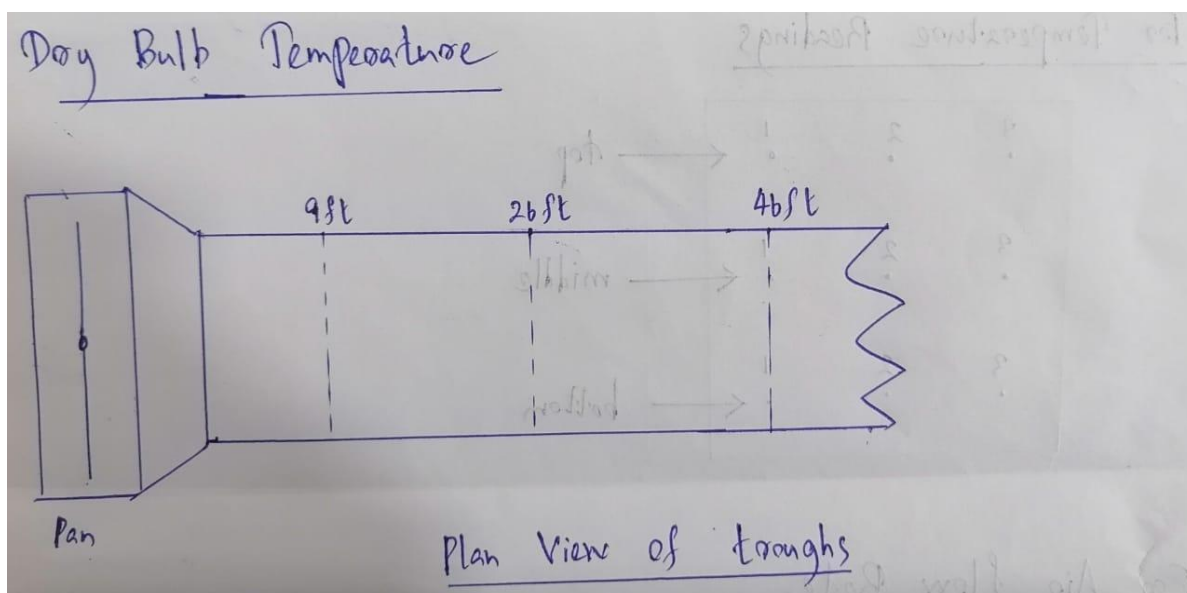


Figure 3: Plan view of the withering trough showing dry-bulb measurement stations (9 ft, 26 ft, 46 ft).

3.4 Total & Static Pressure

Total and static air pressure were measured with a manometer at three stations along the trough (0 ft, 9 ft, and 22 ft from the pan). Readings were converted using: Total pressure (kN/m^2) = $0.1 \times$ (total gauge reading); Static pressure (kN/m^2) = $0.05 \times$ (static gauge reading).

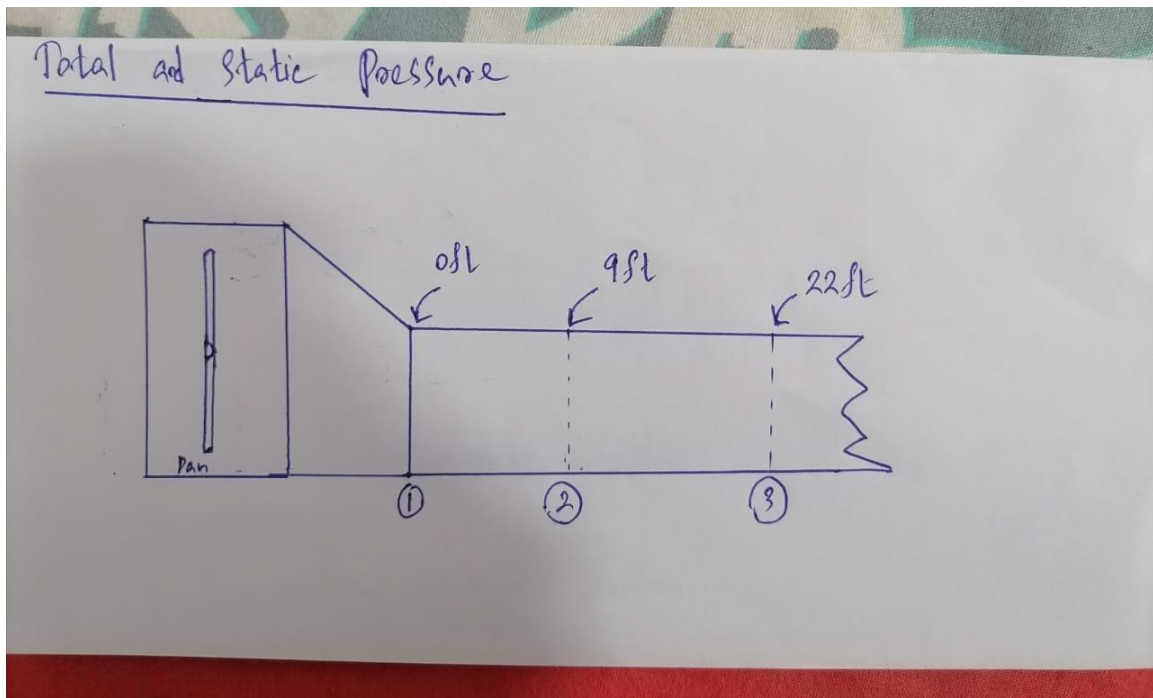


Figure 4: Pressure-measurement stations along the trough length.

4. Data Collected

4.1 Trial Conditions

Five trials were conducted at different cold-air / hot-air damper opening combinations, covering conditions representative of both dry-day and wet-day withering scenarios referenced in the project's control-logic design.

Trial	Time	Wet-Dry Bulb (middle, °F)	Damper Condition
1	6:00 PM	19–20	Cold 100% / Hot 0%
2	7:55 PM	72–79	Cold 100% / Hot 50%
3	9:00 PM	72–80	Cold 50% / Hot 25%
4	9:40 PM	73–79	Cold 100% / Hot 0%
5	10:05 PM	68–71	Cold 100% / Hot 0%

4.2 Trough Temperature Summary (Dry Bulb, °F)

Across all trials and grid positions, dry-bulb readings ranged as follows:

Trough Level	Minimum (°F)	Maximum (°F)
Top	68.9	92.3
Middle	67.4	93.3
Bottom	68.5	91.4

The spread between minimum and maximum readings at each level (up to ~24 °F) indicates a non-uniform temperature distribution across the trough cross-section under several damper settings — consistent with the hotspot behaviour identified in the CFD validation of the existing system.

4.3 Dry-Bulb Temperature Along Trough Length (°F)

Trial	Time	Damper Condition	9 ft	26 ft	46 ft
2	8:35 PM	Cold 100% / Hot 50%	85.0 / 84.4	82.9 / 83.2	82.4 / 82.9
3	9:45 PM	Cold 50% / Hot 25%	81.5 / 80.4	80.3 / 79.6	79.9 / 79.9
4	10:10 PM	Cold 100% / Hot 0%	79.6 / 79.4	78.4 / 78.2	78.5 / 78.4
5	10:30 PM	Cold 100% / Hot 0%	78.3 / 78.0	77.6 / 77.8	77.5 / 77.3

Each cell shows two repeat readings taken at that station.

The maximum recorded temperature (≈ 85 °F) occurs near the mixing/duct entry (9 ft station) and the minimum (≈ 77.4 °F) occurs at the far end of the trough (46 ft) — matching the hotspot findings already reported in the project's mid-term CFD validation.

4.4 Total & Static Pressure (gauge readings)

Trial	Time	Damper Condition	Reading Type	Station 1	Station 2	Station 3
1	6:35 PM	Cold 100% / Hot 0%	Total	0.10	0.90	0.85
			Static	0.28	1.30	1.36
2	8:25 PM	Cold 100% / Hot 50%	Total	0.50	1.30	1.25
			Static	1.20	2.10	2.05
3	9:37 PM	Cold 50% / Hot 25%	Total	0.60	1.15	1.12
			Static	1.05	1.97	2.06
4	10:05 PM	Cold 100% / Hot 0%	Total	0.70	1.45	1.40
			Static	1.40	2.45	2.50
5	10:25 PM	Cold 100% / Hot 0%	Total	1.10	1.43	1.40
			Static	2.20	2.40	2.50

Gauge readings; convert to kN/m^2 via $\text{Total} = 0.1 \times \text{reading}$, $\text{Static} = 0.05 \times \text{reading}$.

Pressure increases from Station 1 (nearest the pan) to Stations 2 and 3 in every trial, indicating resistance build-up along the duct and trough that the mixing-chamber redesign will need to account for.

5. Key Observations

- A clear temperature gradient exists along the trough length, with the highest temperatures near the mixing/duct entry and the lowest at the far boundary — confirming the hotspot identified in the CFD model of the existing system.

- Air-flow readings across the cold-air damper grid were uneven across the 3×3 measurement points, suggesting the existing distribution geometry does not deliver uniform airflow across the trough width.
- Higher hot-air damper openings (e.g., Trial 2: Cold 100% / Hot 50%) produced noticeably higher dry-bulb temperatures than trials with the hot damper closed, confirming the hot/cold air ratio as the primary lever for withering-air temperature control.
- Static and total pressure both rise along the trough length in every trial, showing consistent duct resistance that should be factored into the sizing of the proposed automated mixing and supply system.
- The five damper conditions tested (100/0, 100/50, 50/25) provide representative operating points spanning the dry-day and wet-day withering scenarios used in the project's control-logic design.

6. Relevance to the Project

The data collected during this visit directly supports the ongoing milestones of the WitherMinds project:

- **CFD Validation:** The measured temperature and pressure fields provide the boundary and validation data needed to confirm the CFD model of the existing withering system (Milestone 2).
- **Automated Air-Mixing Design:** The observed non-uniformity in temperature and airflow justifies the proposed redesign of the mixing chamber and distribution geometry to achieve uniform withering conditions.
- **PLC Control Logic:** The real damper-condition/temperature relationships recorded here (e.g., Cold 100%/Hot 50% vs Cold 100%/Hot 0%) provide reference set-points for the PLC-based damper control algorithm under development (Milestone 3).
- **Sensor Placement:** The measurement layouts used at TRI (top/middle/bottom × three lateral positions) inform where temperature and vision sensors should be placed in the automated prototype.

7. Conclusion & Next Steps

The field visit to TRI Talawakele successfully provided a real-world temperature, airflow, and pressure data set from an operating withering trough across five representative damper conditions. This data confirms the existence of temperature hotspots and uneven airflow distribution in the existing system, validating the direction already taken in the project's CFD analysis and justifying the need for an automated hot/ambient air mixing and control system.

Next steps for the WitherMinds team are to:

- Use this data set to finalise validation of the CFD model against the existing trough geometry.
- Incorporate the measured hot/cold damper – temperature relationships into the PLC decision algorithm for automated damper actuation.
- Use the pressure-drop data to size the proposed mixing chamber and supply ducting for the prototype.
- Proceed to prototype fabrication and testing (Milestone 4) informed by these field results.